

Left branch extraction in German fragment answers: Evidence for an in-situ approach

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Abstract This paper investigates the role of contrastivity in Left Branch Extraction (LBE) under ellipsis, presenting experimental evidence from German to support an in-situ approach to clausal ellipsis. While prior research suggests that ellipsis can mitigate island violations, particularly in non-contrastive contexts (Merchant, 2001, 2004), recent proposals argue that ellipsis sites may be syntactically anisomorphic to their antecedents, allowing for alternative derivations that circumvent island constraints (Barros, 2014; Griffiths, 2019). However, these claims largely rely on informally collected acceptability judgments, leaving their empirical robustness in question. To address this gap, controlled acceptability judgment experiments are conducted to test whether contrastivity modulates the acceptability of remnants extracted out of islands. The findings consistently show that contrastively focused remnants force an isomorphic interpretation, leading to significant degradation when the extraction of the remnant violates syntactic constraints. In contrast, anisomorphic readings can be available as a last-resort strategy to avoid island violations. Moreover, a clear difference in acceptability is observed between *wh*-remnants and non-*wh* remnants, indicating that the remnant's ability to move determines its island sensitivity. These results underscore the necessity of an in-situ approach to clausal ellipsis and provide evidence against a general island repair effect in LBE contexts. This paper argues that (i) ellipsis can occur 'around' a designated constituent, (ii) syntactic islands are not repaired by ellipsis, (iii) the ellipsis site need not be syntactically isomorphic to its antecedent if independent conditions are met, (iv) remnants A'-move in elliptical configurations only if they would A'-move if ellipsis had not applied, and (v) contrastive focus on the remnant forces an isomorphic interpretation of the elliptical clause's syntax, without enforcing any restrictions on presentational focus.

Keywords: clausal ellipsis; island sensitivity; left branch extraction; German; acceptability judgments

1. Introduction

The study of islands and clausal ellipsis has played a central role in understanding syntactic constraints and the boundaries of grammaticality. In the literature which assumes that ellipsis clauses are derived from full underlying clauses (e.g. Ross, 1969; Merchant, 2001), the structure in the ellipsis site plays a crucial role in explaining island violations. Configurations such as (1) have been at the center of this debate. Several approaches in the literature aim to explain why (1) is acceptable, although its non-elliptical counterpart is ungrammatical, as it includes ungrammatical extraction out of the left branch (an island position; Ross, 1969), as shown in (2).

- (1) A: Lena married a tall man.
B: Yes, very tall.
(2) B: * Yes, [very tall]₁ she married [_{island} a t₁ man].

It has been proposed in the literature that the structure in the ellipsis site does not need to be syntactically parallel (Vicente 2008; Rodrigues et al., 2009; Szczegielniak, 2005, 2008; van Craenenbroeck, 2010; Barros 2014). If an anisomorphic ellipsis site is available, the fragment answer (the *remnant*) (1B) could be derived from a copular clause such as (3), where struck-through text represents ellipsis.

- (3) B: Yes, [very tall]₁ ~~he is~~ t₁. (adapted from Barros, 2016: 4-5)

Crucially, an anisomorphic ellipsis site does not include an island and could therefore straightforwardly account for the acceptability of (1). However, similar configurations such as (4) are deemed unacceptable.

- (4) A: Lena married a tall man.
B: * No, short.

(adapted from Merchant, 2004: 688)

To capture (3)'s unacceptability, Barros (2014) and Griffiths (2019) propose that the presence of contrastive focus forces one to interpret the ellipsis as isomorphic to its antecedent, which yields an island effect if the remnant's correlate in the antecedent is itself island-bound, as is the case in (4). Much of the literature debating the structure in the ellipsis site has centered on English and relies on informally collected acceptability judgments. This is not ideal, because it means that the debate currently rests on subtle lexical-semantic comparisons (see e.g. the renowned 'hard worker' example in Barros 2014: 31) for which informal methods of data collection are unreliable. The purpose of this paper is to address this issue by employing controlled acceptability judgment experiments and by examining the difference between configurations such as (2) and (3) in a language that exhibits a morphological distinction between attributive and predicative adjectives, namely German.

This research contributes to the growing body of work that seeks to clarify the nature of island effects. In particular, the emerging consensus in the literature is that only left branch islands are genuinely syntactic in nature. This position has been recently supported by Lu et al.'s (2024) meta-analysis of satiation studies on islands and Culicover et al.'s (2022) investigation into the factors for the unacceptability of island violations. This paper offers further empirical support for this view, reinforcing the distinction between syntactic and non-syntactic island effects.

In addition to addressing the nature of islands, the study provides evidence for the in-situ approach to clausal ellipsis. The results support this theoretical framework while casting doubt on earlier influential proposals. The in-situ approach posits that ellipsis operates selectively, i.e. only non-focused elements are deleted, while focused elements survive ellipsis (e.g. Abe, 2015; Griffiths, 2019; Stigliano, 2022). This approach has several implications: (i) island violations are not repaired under ellipsis, (ii) remnants undergo A'-movement in elliptical configurations if they would do so in non-elliptical configurations, and (iii) the presence of contrastive focus on the remnant forces one to interpret the ellipsis site as syntactically parallel to its antecedent. The research questions center on testing these predictions and examining whether the in-situ approach offers a more accurate account of island effects under ellipsis than alternative approaches.

The significance of this work lies in its challenge to earlier foundational proposals, such as those put forth by Merchant (2001) and Lasnik (2001). The results diverge from previous experimental studies, including Yoshida et al. (2019), which observed island sensitivity in ellipsis contexts. This paper explores possible reasons for this misalignment, suggesting that differences in the types of island-crossing movement investigated may account for the discrepancies.

Several of the proposed claims challenge earlier influential proposals, such as Merchant (2001, 2004), who argues that remnants must A'-move to escape an ellipsis operation applying unselectively to a clause-sized constituent, and Lasnik (2001), who proposes that elliptical clauses and their antecedents are always syntactically parallel and that islands are repaired by ellipsis.

The structure of the paper is as follows: Section 2 offers a detailed overview of the theoretical background, with a focus on why the left branch island stands out from other islands and how it behaves under ellipsis. Section 3 reports on the acceptability judgment experiment. Section 4 revisits the hypotheses and discusses the implications of the findings, arguing that the experimental evidence supports the in-situ approach to ellipsis. Finally, Section 5 concludes the paper with a summary of the contributions.

2. Theoretical background

This section provides a comprehensive examination of the theoretical background, outlining the concept of islands and differentiating between syntactic islands and those stemming from pro-

cessing constraints. The section focuses on island effects under ellipsis and how various theoretical perspectives address this phenomenon. Moreover, the role of contrastive focus in causing island sensitivity under ellipsis is highlighted. Finally, previous experimental studies addressing these research questions are summarized.

2.1 Theoretical approaches to islands in non-elliptical configurations

There are several major theoretical perspectives on island effects. Grammatical accounts argue that island phenomena arise from structural principles such as movement, locality constraints, or phase theory (Ross, 1967; Chomsky, 1986; Phillips, 2006; Abels, 2012). In contrast, processing-based accounts attribute island effects to cognitive limitations rather than syntactic rules, e.g., memory load, working memory constraints, and discourse factors (Pritchett, 1991; Kluender & Kutas, 1993; Hofmeister & Sag, 2010).¹

The tension between grammatical and processing-based accounts of island effects has been the subject of extensive experimental investigation. Kluender and Kutas (1993) found that certain island violations do not trigger the same neural responses as unambiguous grammatical violations, pointing to processing constraints. Hofmeister and Sag (2010) demonstrated that factors such as semantic richness and discourse cues can improve acceptability, supporting a processing-based explanation. Goodall (2015) provided further evidence that processing effects influence acceptability but also concluded that these effects are likely to operate independently of the grammatical constraints responsible for island effects. Sprouse and Villata (2021) provide a comprehensive overview of experimental findings, emphasizing how empirical methods can help disentangle grammatical and processing influences. Culicover and Winkler (2022) and Culicover et al. (2022) also contribute to this debate, arguing based on experimental and theoretical evidence that many island effects reflect processing constraints rather than grammatical rules. Finally, Yoshida et al.'s (2014) self-paced reading experiment showed that island effects cannot be fully explained by processing constraints alone. Instead, they must at least partly stem from grammatical constraints. The consensus in the literature is that the unacceptability of certain configurations may be attributed to different constraints. However, the prevailing view in the literature is that strong islands such as the left branch condition are grammatical constraints. This is outlined below.

2.2 The left branch condition as a strong island

The Left Branch Condition (LBC) is an island constraint which disallows the extraction of the leftmost NP of a larger NP (Ross, 1967). In the literature, this constraint has been generalized for any prenominal modifier, including attributive adjective modifiers (Barros et al., 2015), as shown in (5).

- (5) a. *Whose_i did you see [t_i father]?
 b. *Which_i did you buy [t_i car]?
 c. *That_i he saw [t_i car].
 d. *Beautiful_i he saw [t_i houses].
 e. *[How much]_i did she earn [t_i money]? (Bošković, 2005: 2)

The LBC is often considered a genuinely syntactic constraint, in contrast to other island effects that may arise from processing difficulties. While many island constraints, e.g., complex NP islands or wh-islands, have been argued to reflect processing limitations or pragmatic factors,

¹ In this section, the focus lies on the two ends of the spectrum, i.e., grammatical and processing-based accounts. There are additional approaches that incorporate functional or discourse-based factors. However, including such accounts would require a more detailed engagement with theories of discourse structure, which goes beyond the scope of the paper.

the LBC is distinct in exhibiting consistent grammatical behavior. The status of the LBC as a grammatical constraint is apparent by (i) its crosslinguistic variation, (ii) the very low acceptability scores, and (iii) the lack of a satiation effect. Each property of the LBC is outlined below.

Although the LBC is active in languages such as English, German, French, Danish, Italian, and Finnish (Ross, 1967), it is not universal. Certain languages such as Latin (Ross, 1967) and many Slavic languages (Abels, 2003; Bošković, 2005, 2008, 2012) permit Left Branch Extraction (LBE). Empirical data from languages such as Serbo-Croatian and Latin that allow LBE (ibid.) support the categorization of the LBC as a grammatical island constraint, which is subject to parametric crosslinguistic differences, instead of processing constraints. To investigate island sensitivity under ellipsis, this paper focuses on German, i.e., a language that prohibits LBE in non-elliptical constructions.

Island violations do not pattern uniformly. For instance, Hofmeister and Sag (2010) find that complex NP islands, *wh*-islands, and adjunct islands become significantly more acceptable if more informative elements (which-NP versus *what/who*) are used, easing the processing burden. Moreover, Kush et al. (2018) find significant differences across speakers' judgment of whether-islands in Norwegian. Nyvad et al. (2022) observe that providing more context significantly reduces the violation associated with adjunct extraction in English. Crucially, claims that island effects are generally reducible to discourse or processing factors remain least substantiated for the LBC.

Recent studies have explored whether LBC violations exhibit satiation effects, i.e. whether repeated exposure leads to increased acceptability of island violations (Snyder, 2022). If such effects are found, this would provide evidence that the island is a processing constraint and not grammatical in nature. However, findings suggest that LBC violations are comparatively resistant to satiation, especially when contrasted with other island types such as *wh*-islands and complex NP islands (Culicover et al., 2022). In a comprehensive meta-analysis of satiation effects, Lu et al. (2024) concluded that the LBC behaves like a genuinely syntactic island, with its resistance to satiation serving as further evidence of its grammatical origin. These findings align with earlier observations by Snyder (2000) and Sprouse (2009), who noted that not all island constraints are equally prone to frequency or exposure-based effects. In particular, the unacceptability of LBE remains robust across repeated exposure. Taken together, this body of work supports the view that the LBC reflects a grammatical constraint rather than a processing constraint.

2.3 Approaches to clausal ellipsis

To account for the (un)acceptability of elliptical clauses, several approaches have been proposed in the literature. The two main theoretical frameworks to clausal ellipsis differ in whether they assume syntactic structure in the ellipsis site (sententialist approaches) or not (non-sententialist approaches).

Non-sententialist approaches assume that elliptical clauses are not derived from underlying clauses. Instead, they are licensed directly by the discourse (e.g., Ginzburg & Sag, 2000; Culicover & Jackendoff, 2005; Sag & Nykiel, 2011). Because the derivation involves no syntactic movement, this view inherently predicts that all elliptical remnants should be island insensitive. Consequently, non-sententialists can straightforwardly explain why certain elliptical clauses are acceptable even when their non-elliptical counterparts trigger island violations. Because this approach rejects any covert syntactic material in the ellipsis site, the problematic notion of ellipsis repair is rendered entirely unnecessary.

However, configurations showing island sensitivity present a challenge for the non-sententialist approach. Because the non-sententialist approach assumes a uniform, structureless derivation, it incorrectly predicts contrastive *wh*-remnants to be completely acceptable. To better investigate how island sensitivity, licensing, and interpretation are regulated, this paper focuses on the sententialist approach to clausal ellipsis.

Within the sententialist view, the ongoing debate centers on five critical questions: (i) the internal structure of the ellipsis site, (ii) the selectivity of the ellipsis operation, (iii) the availability

and nature of movement under ellipsis, (iv) the availability and mechanism of ellipsis repair, and (v) the role of non-syntactic factors in ellipsis licensing. In the following, I outline each point of discussion in more detail.

Early work established that ellipsis interpretation is facilitated by parallelism (isomorphism) between the antecedent and the elliptical clause (e.g. Carlson, 2001). However, there is no consensus in the literature whether strict syntactic isomorphism is required or not (see e.g. Fox & Lasnik, 2003; Takahashi & Fox, 2005 for arguments in favor of strict parallelism), as in (6).

- (6) A: Lena married someone.
 B: Yes, [Paul]₁ ~~she married~~ _{t₁}. (isomorphic)
 B': Yes, [Paul]₁ ~~that was~~ _{t₁}. (anisomorphic)

If anisomorphic ellipsis sites are possible, a second question arises: when are they available? Some researchers argue that they are generally available (e.g. Rodrigues et al., 2009; Szczegielniak, 2008), whereas others treat them as a last-resort strategy (e.g. Barros, 2014). This distinction is crucial for the current discussion because anisomorphic ellipsis sites often avoid island violations entirely, as the ellipsis site no longer contains an island. That is, no island sensitivity is predicted under an anisomorphic resolution.

The second fundamental question addresses the mechanism of the ellipsis operations. Following Merchant's (2001) influential work, the dominant assumption in the literature is that ellipsis operates unselectively. That is, ellipsis is assumed to delete an entire syntactic constituent (e.g., TP). However, more recent approaches to clausal ellipsis have argued that ellipsis is selective, i.e. only non-focused elements are deleted (e.g. Reich, 2007; Ott & Struckmeier, 2018), as shown in the example with multiple *wh*-remnants in (7), where focused elements are capitalized. This selectivity is a prerequisite for in-situ approaches, which are discussed below.

- (7) John went to the movies, but I don't know WHO ~~he went to the movies~~ WITH _{t₁}.
 (Ott & Struckmeier, 2018: 400)

The third disagreement concerns the availability of movement under ellipsis, as shown in (8). In his move-and-delete (M&D) approach, Merchant (2001) argues that all remnants must move to escape ellipsis. This requires stipulating exceptional, ellipsis-driven movement for phrases that are otherwise immobile in non-elliptical configurations such as the non-*wh* remnant, as in (B).

- (8) A: Lena married a tall man.
 B: Yes, [very tall]₁ ~~she married~~ [_{island} ~~a~~ _{t₁} ~~man~~]. (movement approach)
 B': Yes, ~~she married~~ [_{island} ~~a~~ [very tall] ~~man~~]. (in-situ approach)

In contrast, in-situ approaches reject the idea of ellipsis-driven movement. These approaches differ in whether they assume that all remnants stay in their base position (e.g. Abe, 2015) or only remnants that would move in non-elliptical configurations also move under ellipsis (e.g. Griffiths, 2019), as in (B'). Crucially, if a remnant stays in-situ and therefore does not cross an island boundary, no island sensitivity is predicted.

To account for the high acceptability of elliptical strings whose non-elliptical counterparts are ungrammatical, two competing explanations exist: Supporters of ellipsis repair argue that syntactic violations such as island violations are genuinely generated by movement but are subsequently repaired by the phonological deletion of the ungrammatical structure (e.g. Lasnik, 2001; Merchant, 2008; Rottman & Yoshida, 2013; Potter, 2017). In contrast, other approaches reject repair effects altogether and instead attribute the acceptability to the availability of underlyingly grammatical, anisomorphic ellipsis resolutions that do not contain an island (e.g., Barros, 2014; Griffiths, 2019).

Finally, sententialist approaches differ in whether ellipsis licensing is confined to syntactic constraints or whether non-syntactic factors influence ellipsis licensing. Merchant (2001, 2004) treats ellipsis licensing as a purely syntactic operation. Alternative approaches argue that

ellipsis licensing involves non-syntactic constraints such as focus (e.g. Reich, 2007), remnant type (Ott & Struckmeier, 2018; Griffiths, 2019), and referent selection (Schiele, 2025).

The discussion above shows that competing theories differ primarily in their predictions about island sensitivity. The present study revisits these predictions by isolating grammatical island constraints from processing-related influences, arguing that previous experimental paradigms often conflated the two.

2.4 Contrastive focus and its effect on ellipsis interpretation

Expressions that add new information to a discourse such as the fragment answer in (9) bear focus (Schwarzschild, 1999).

- (9) A: Who stole the cookie?
B: [PETER]_F stole the cookie. (adapted from Krifka, 2008: 250)

Focus marks the presence of discourse alternatives (Krifka, 2008). For the purpose of this paper, the set of alternatives consists of the available referents in the discourse. Two types of focus are distinguished in the literature: presentational and contrastive focus (first discussed in Halliday, 1967; see Kiss, 1998 for an overview). In this paper, presentational focus refers to focus placed on elements that share their referent with their correlate, as in (9). In contrast, elements which introduce a new referent, often used in corrections such as (10), bear contrastive focus (following Barros, 2014; Abels, 2019; Palaz et al., 2025).

- (10) A: Mary stole the cookie.
B: (No,) [PETER]_F stole the cookie. (Krifka, 2008: 250)

Although accounts of ellipsis licensing differ in their treatment of the semantic component (question-based; Ginzburg & Sag, 2000, AnderBois, 2011; Weir, 2014 versus focus-based; Schwarzschild, 1999; Merchant, 2001, 2004; Barros & Kotek, 2019), there is broad agreement that focus structure affects ellipsis interpretation. In particular, a robust finding in the literature is that island sensitivity is primarily associated with contrastive focus (Merchant, 2008; Barros, 2014; Griffiths & Lipták, 2014; Palaz, 2023). Presentational remnants are often acceptable despite their island-bound correlate, whereas contrastive remnants typically degrade in the same environment, as shown in (11)-(14).

- (11) A: [_{island} The rumour that one of the teaching assistants was fired] is true.
B: Yeah, JUDY. (Griffiths, 2019: 12)
- (12) A: [_{island} The rumour that SALLY was fired] is true.
B: * No, JUDY. (ibid.)
- (13) Abby wants to hire someone who speaks a Balkan language, but I don't remember which.
(adapted from Griffiths & Lipták, 2014: 207)
- (14) * Abby wants to hire someone who speaks GREEK fluently, but I don't remember what OTHER language. (ibid.)

It remains an open question why island sensitivity is restricted to contrastive remnants. Three main schools have been proposed in the literature. The first line of analysis derives the contrast from movement. Winkler (2005, 2013) and Gengel (2013) argue that contrastive remnants undergo focus movement, which is subject to island constraints. A second approach is developed by Griffiths and Lipták (2014), who argue that the distinction follows from differences in the interaction between contrastive focus and ellipsis repair. A third line of work attributes island-sensitivity to the availability of an anisomorphic ellipsis site. Barros (2014) argues that presentational remnants permit an anisomorphic interpretation that avoids the island, whereas contrastive remnants require an isomorphic interpretation that preserves the violation. Griffiths

(2019) develops this view further by proposing that non-wh remnants may also remain in situ in wh-languages, which restricts island-sensitivity to contrastive wh-remnants.

2.5 Previous research on island sensitivity under clausal ellipsis

A growing experimental literature has investigated whether clausal ellipsis obviates island violations, often with the explicit aim of testing competing approaches to clausal ellipsis. While this body of work has yielded influential results, a closer examination reveals conflicting results as well as systematic methodological oversights that complicate their interpretation. In particular, many studies do not adequately control the availability of short or anisomorphic ellipsis sources, making it difficult to determine whether apparent island insensitivity reflects genuine repair or merely island evasion. This section reviews the relevant experimental findings and argues that these shortcomings motivate a renewed focus on LBE, which provides a more constrained testing ground for island sensitivity under ellipsis.

Schmeh et al. (2015) test the acceptability of island-violating fragments answers and find a difference in acceptability between fragment answers that supplement new information to the discourse and those correcting the preceding utterance (compare (15) and (16)).

- (15) A: John met a guy [_{island} who speaks a very unusual language].
B: Yes, Albanian. (Schmeh et al., 2015: 199)
- (16) A: Does Abby speak the same Balkan language [_{island} that Ben speaks].
B: * No, Charlie. (ibid.: 200)

The authors argue that the fragment answer in (16) is degraded, as corrective fragments are more difficult to process than their supplementing counterparts. They therefore attribute the unacceptability of corrective fragments to non-syntactic factors.

Similarly, Potter (2017) tests island sensitivity under ellipsis in English in a series of acceptability judgment experiments. He finds that fragments whose correlates occur inside relative clause (RC) islands were judged to be broadly acceptable. For corrective cases such as (17), he observes the fragments to be “partially sensitive” (2017: 102) to island effects.

- (17) A: Jordan mocked the salesman [_{RC} who sold a car to an elderly customer].
B: No, to Ashley. (adapted from Potter, 2017: 97)

Similarly, Yoshida et al. (2019) find that fragment answers such as (18) were only minimally degraded compared to the baseline condition without an island, as in (19).

- (18) A: James met the journalist [_{RC} who got a commissioned piece].
B: No, a salaried position. (adapted from Yoshida et al., 2019: 1521)
- (19) A: James heard that the journalist got a commissioned piece.
B: No, a salaried position. (ibid.)

Based on these results, the authors conclude that contrastive ellipsis is island insensitive and use their results as confirmation for an island repair via ellipsis. However, there are reasons to believe this conclusion is premature. First, Potter (2017) and Yoshida et al. (2019) only tested definite relative clause islands. To generalize for clausal ellipsis, different types of islands need to be tested, as island effects are known to differ depending on island type, such that grammatical restrictions are referred to as strong islands, whereas non-syntactic factors such as processing burdens are referred to as weak islands (see Szabolcsi, 2006 for an overview). Although extraction out of subjects is supposedly ruled out by the syntax and therefore unacceptable, the effects of subject islands vary across languages (Sprouse et al., 2016), and, contrary to predictions in the literature, extraction out of subjects was found to be acceptable (Hofmeister & Sag, 2010; Chaves & Dery, 2019; Abeillé et al., 2020; Palaz, et al. 2025), suggesting that extraction out of subjects is unsuitable to test islands effects. Second, Wu (2022) identifies several methodological concerns in Yoshida et al.’s (2019) study, including issues related to the use of stripping constructions, the manipulation of contrastive focus, and a finality confound in the experimental stimuli. Third,

fragments with utterance-final correlates pattern exceptionally, as they are consistently judged to be acceptable, as noted by Griffiths and Lipták (2014: fn10), Barros et al. (2015), and Reeve (2016). Because this confound is not controlled in the stimuli, the observed acceptability of island-bound fragments cannot be straightforwardly attributed to island insensitivity under ellipsis. That is, the conclusion that the experimental results in Potter (2017) and Yoshida et al. (2019) suggest island repair appears unwarranted.

A strong empirical support for the ‘island evasion’-approach comes from Palaz et al.’s (2025) experimental work on *wh*-remnants. In a forced-choice task with three alternative completions, participants were more likely to choose a completion that is structurally parallel to the antecedent clause for contrastive remnants, as in (21), compared to presentational remnants, as in (20). This suggests that (i) anisomorphic clauses are available and preferred as an ellipsis site under appropriate discourse conditions and (ii) contrast forces one to interpret ellipsis site as isomorphic to the antecedent clause, as suggested by Barros (2014) and Griffiths (2019).

(20) Captain America amazed Mike, but his mother cannot recall which other superhero...
(Palaz et al., 2025: 9)

(21) A superhero amazed Mike, but his mother cannot recall which superhero...
(ibid.)

These results provide direct empirical support for Barros’s proposal that contrast enforces an isomorphic interpretation of the ellipsis site, thereby blocking island evasion strategies. Importantly, when such strategies are unavailable, island sensitivity re-emerges.

Further evidence for an evasion-based account is provided by Wu (2022), who argues that apparent island violations can be avoided through the use of anisomorphic ellipsis sites. She discusses examples like (22), where the ellipsis site could be interpreted as isomorphic (a) or anisomorphic (b) to the antecedent clause, as shown in (23).

(22) They hired someone who speaks French yesterday, and also German. (Wu, 2022: 5)

(23) a. ...and also German_i ~~they hired~~ [_{RC} ~~someone who speaks t_i yesterday~~].
b. ...and also German_i ~~she speaks t_i~~. (ibid.)

Crucially, the truth values of the isomorphic and anisomorphic reading differ. In the reading (a), a German speaker and a French speaker were hired, whereas one bilingual was hired in the reading (b). Wu reports that all seven consultants rated (22) as fully acceptable and, when asking how many people were hired, all of her consultants reported that only one bilingual was hired, indicating that the anisomorphic ellipsis site was available for all consultants. Wu’s findings, therefore, independently support Barros’ (2014) claim that apparent island insensitivity under ellipsis often reflects structural evasion rather than genuine repair of island violations.

One might question the generalizability of Wu’s findings, as her results are based on the judgments of only seven speakers. However, Cortés Rodríguez and Griffiths (2024a,b) provide further experimental evidence from German, showing that anisomorphic ellipsis sites are available. In an acceptability judgement experiment, 31 speakers rated the acceptability of elliptical clauses such as (24). Crucially, the dative marking in (a) forces one to interpret the ellipsis site as isomorphic to the antecedent, whereas the nominative marking in (b) is only compatible with an anisomorphic interpretation of the ellipsis site.

(24) Nadine hat einem Investor erzählt, dass er Tanja einem
Nadine has a.DAT investor told that he Tanja a.DAT
Verkäufer nahelegen soll, aber ich weiß nicht genau,
clerk recommend should but I know not exactly
‘Nadine told an investor that he should recommend Tanja to a clerk, but I don’t know...’
a. **welch-em** Investor **welch-em** Verkäufer.
which-DAT investor **which-DAT** clerk
Interpretation: ‘...to [which investor]_i Nadine said that he_i should recommend Tanja to which clerk.’

- (27) A: Lena hat einen groß-en Mann geheiratet.
 Lena has a tall-ACC man married
 ‘Lena married a tall man.’
- B: Ja, [sehr groß-en]₁ ~~Lena hat einen~~ ~~t₁~~ ~~Mann~~ ~~geheiratet~~.
 Yes very tall-ACC Lena has a man married
 ‘Yes, very tall (Lena married a man).’
- B’: Ja, [sehr groß]₁ ~~er~~ ~~ist~~ ~~t₁~~.
 Yes very tall.NOM he is
 ‘Yes, very tall (he is).’

3.2 Motivation and hypotheses

The present experiments address several gaps in the literature by testing LBE under clausal ellipsis in German. Specifically, they compare two types of remnants: *wh*-remnants and non-*wh* remnants. This design allows for a direct test of the competing predictions of movement-based and in-situ approaches to ellipsis. The experiments also provide the first controlled investigation of LBE under ellipsis in German.

The study contributes methodologically in four respects. First, it examines a strong island constraint that has not previously been investigated experimentally in German. Second, it uses morphological cues to sharply restrict the availability of anisomorphic ellipsis sources, reducing a major source of ambiguity in the interpretation of ellipsis data. Third, it manipulates contrastive focus in order to test whether conditions that favor isomorphic interpretations reinstate island sensitivity. Finally, it provides quantitative evidence for a domain in which theoretical claims have relied primarily on introspective judgments.

A central motivation for focusing on LBE is that previous experimental work on ellipsis and islands has concentrated largely on relative clause islands, *wh*-islands, and other constructions whose status as grammatical constraints has been questioned. In contrast, the LBC is widely regarded as one of the strongest candidates for a genuinely syntactic island constraint. As a result, LBE provides a particularly stringent test case for theories of ellipsis. If ellipsis genuinely repairs island violations, repair effects should be observable even in the context of LBE. Conversely, persistent degradation despite ellipsis would constitute especially strong evidence against repair-based approaches.

The comparison between *wh*-remnants and non-*wh* remnants allows the competing theoretical accounts to be distinguished. If neither remnant type exhibits island sensitivity, this outcome would be compatible either with genuine island repair under ellipsis or with analyses in which remnants are generated in situ and therefore do not involve island-crossing movement. If island effects emerge for one or both remnant types, however, this would indicate that movement under ellipsis is possible and that ellipsis does not systematically repair island violations.

The experiments also provide a direct test of repair-based and evasion-based accounts of apparent island insensitivity. Under repair approaches, LBE should remain acceptable even when alternative ellipsis sources are unavailable and contrastive focus favors an isomorphic interpretation. Under evasion approaches, by contrast, apparent island insensitivity arises because speakers adopt non-isomorphic derivations that avoid the island violation. Consequently, island effects are expected to re-emerge when morphological and discourse cues restrict access to such alternative sources and favor an isomorphic interpretation.

The experiments therefore test the following predictions. First, if ellipsis genuinely repairs island violations, no substantial island effect is expected for either *wh*-remnants or non-*wh* remnants, regardless of the availability of alternative ellipsis sources. Second, if both remnant types involve movement, island effects should be observed for both *wh*-remnants and non-*wh* remnants. If only one remnant type undergoes movement, island effects should be restricted to that remnant type. Third, if apparent island insensitivity reflects evasion rather than repair, island effects should emerge when morphological and discourse cues favor an isomorphic interpretation

and restrict access to non-isomorphic alternatives. Finally, if neither remnant type involves movement, neither remnant type is expected to exhibit island sensitivity. Together, these predictions allow the experiments to distinguish between repair-based, evasion-based, movement-based, and in-situ approaches to clausal ellipsis.

3.3 Participants

Participants were recruited and remunerated via Prolific (www.prolific.com). All participants were over 18 years old and German monolinguals. Participants were unfamiliar with the purpose of the experiment. Monolingualism was controlled by filtering participants on Prolific and asking participants about their native language(s) before and after the experiment. Participants were removed from the analysis (i) if they were exposed to a second language before the age of 6, (ii) if they failed two or more of the attention checks, or (iii) if their responses to the standardized fillers were more than two standard deviations away from the mean.

32 German monolinguals were recruited in each experiment. In Experiment 1, 2 bilinguals were identified and removed from the analysis. 1 participant failed the attention checks. 2 more participants were removed as outliers based on their responses to the standardized fillers. The final sample consisted of 27 participants (8 females, 19 males), aged between 19 and 52 years with a mean age of 29.4 years.

In Experiment 2, 4 bilinguals were removed and 1 participant was removed as outlier based on their rating of the standardized fillers. All participants passed the attention checks. The final sample included 27 participants (9 females, 18 males), aged between 19 and 71 years with a mean age of 36.1 years.

3.4 Materials

Experiment 1 tested wh-remnants, as in (28), while Experiment 2 featured non-wh remnants, as in (29).

- (28) A: Das neue Gesetz wird vor allem reichen Bürgern zugutekommen.
 The new law will above all rich.ACC citizens benefit
 ‘The new law will mostly benefit rich citizens.’
 B: Wie reich-en? (marked)
 how rich-ACC
 ‘How rich?’
 B’: Wie reich? (unmarked)
 how rich.NOM
 ‘How rich?’
- (29) A: Das neue Gesetz wird vor allem reich-en Bürgern zugutekommen.
 The new law will above all rich-ACC citizens benefit
 ‘The new law will mostly benefit rich citizens.’
 B: Ja, sehr reich-en. (marked, presentational)
 yes very rich-ACC
 ‘Yes, very rich.’
 B’: Ja, sehr reich. (unmarked, presentational)
 yes very rich.NOM
 ‘Yes, very rich.’
 B’’: Nein, arm-en. (marked, contrastive)
 no poor-ACC
 ‘No, poor.’

B''': Nein, arm. (unmarked, contrastive)
no poor.NOM
‘No, poor.’

In both experiments, stimuli were distributed across 12 Latin-squared lists, whereby each participant saw only one repetition of each lexical set. In addition, each list included 3 attention checks and standardized fillers. The fillers were adapted from Gerbrich et al. (2019) and represented each level on the acceptability scale. Both experiments utilized a 2:1 ratio of fillers to test items. In Experiment 1, participants saw 3 repetitions per condition, yielding 81 observations per condition. In Experiment 2, participants saw 6 repetitions per condition, resulting in 162 observations per condition.

Moreover, Experiment 2 controlled embedding to check whether embedded remnants can be compared to fragment answers. That is, one third of the items utilized one-sentence stimuli, as in (30), one third included a dialogue format with embedded remnants, as in (31), and one third employed a dialogue format without embedding, as in (32).

- (30) Das neue Gesetz wird vor allem reichen Bürgern zugutekommen, aber ich weiß nicht wie reich.
‘The new law will mostly benefit rich citizens, but I don’t know how rich.’
- (31) A: Das neue Gesetz wird vor allem reichen Bürgern zugutekommen.
‘The new law will mostly benefit rich citizens.’
B: Weißt du wie reich?
‘Do you know how rich?’
- (32) A: Das neue Gesetz wird vor allem reichen Bürgern zugutekommen.
‘The new law will mostly benefit rich citizens.’
B: Wie reich?
‘How rich?’

In the literature, the terminology for sluicing and fragment answers are not clearly defined. Crucially, the difference in embedding was found to not be a significant confound, indicating clausal ellipsis in embedded configurations patterns similarly to clausal ellipsis in dialogues. I will therefore merely distinguish between wh- and non-wh remnants, rather than sluices and fragment answers. Moreover, the difference between passive and active sentence or with animate and inanimate referents did not yield significant differences.

3.5 Method

The experimental design was identical in both experiments. Both experiments employed a within-subject factorial design. Participants were asked to rate the naturalness of the presented stimuli on a 7-point scale, where 1 represented ‘fully unnatural’ and 7 represented ‘fully natural’.

Before the experiments, participants were informed about consent and data protection. After a practice trial, participants were presented with the critical items, filler items, and three attention checks. In addition, participants filled out a survey about their socio-demographic background. Both experiments were hosted by Gorilla Experiment Builder (Anwyl-Irvine et al., 2018). Data collection took place in July 2024.

Experiment 1 tested the factor CLAUSE (isomorphic or anisomorphic) for wh-remnants. Experiment 2 crossed the factors CLAUSE (isomorphic and anisomorphic) and FOCUS (presentational or contrastive) for non-wh remnants.

3.6 Data analysis

The data were analyzed using cumulative link mixed-effects models (CLMMs), implemented in the ordinal package in R (Christensen, 2019; R Core Team, 2022). The dependent variable was

the ordinal acceptability score. Fixed effects included the factors of each experiment, i.e. CLAUSE in Experiment 1 and CLAUSE and FOCUS in Experiment 2. The categorical predictors were treatment-coded with the isomorphic-condition as reference level in Experiment 1 and the isomorphic-condition and presentational-condition as reference levels in Experiment 2.

To determine an appropriate random-effects structure, a set of candidate models was fitted that all shared the same fixed-effects structure but differed in their random-effects specification. Specifically, models varied in whether they included random intercepts only or random intercepts plus random slopes for each fixed effect, and their interaction, for participants and lexical sets. This resulted in a sequence of models ranging from a maximal random-effects structure (including correlated random slopes for both predictors and their interaction for both grouping factors) to a minimal structure including only random intercepts. Model comparison was carried out using the Akaike Information Criterion (AIC). The model with the lowest AIC value was selected as the best-fitting model among the candidates.

3.7 Results

Participants' ratings across test conditions in both experiments are shown in Figure 1.

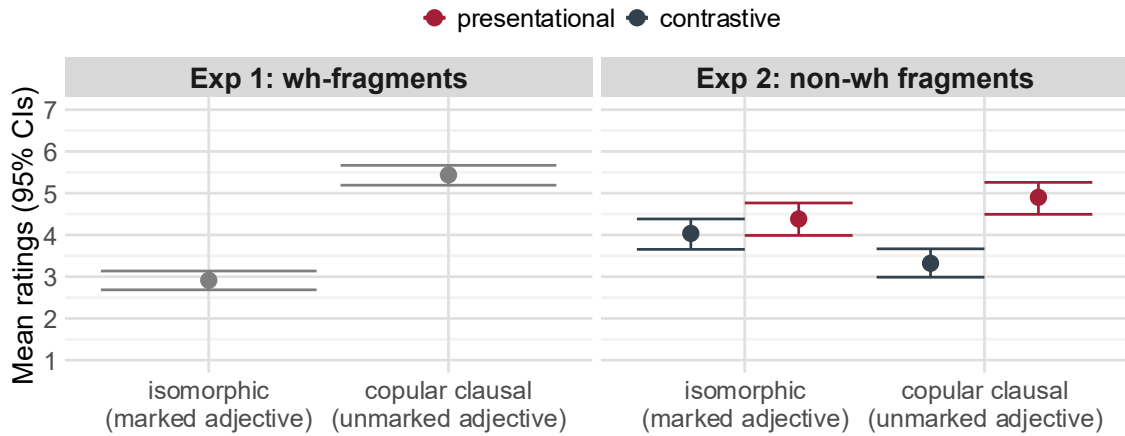


Figure 1: Mean ratings across conditions in Experiment 1 and 2.

The final CLMM in Experiment 1 included CLAUSE as fixed effect with random slopes and random intercepts for CLAUSE by participant and lexical set. The effect of CLAUSE is highly significant ($\beta = 4.28$, $SE = 0.54$, $z = 7.89$, $p < 0.01$). That is, unmarked wh-remnants are preferred to marked wh-remnants.

In Experiment 2, the final CLMM included CLAUSE, FOCUS, and their interaction as fixed effects with random intercepts for participants and lexical sets as well as random slopes for CLAUSE by participant and lexical set. The effect CLAUSE did not reach significance ($\beta = -0.72$, $SE = 0.71$, $z = -1.01$, $p = 0.31$). However, the effect of FOCUS ($\beta = -2.37$, $SE = 0.33$, $z = -7.28$, $p < 0.01$) and the interaction between CLAUSE and FOCUS ($\beta = 1.82$, $SE = 0.44$, $z = 4.15$, $p < 0.01$) were significant, with [unmarked, contrastive] items receiving the lowest ratings, as predicted. Crucially, the effect of CLAUSE is significant in Experiment 1, but not Experiment 2.

4. General discussion

This section evaluates the research questions based on the observed pattern. Moreover, I argue that an in-situ approach can straightforwardly account for the data, while the observed pattern poses challenges for alternative approaches to clausal ellipsis.

4.1 Revisiting hypotheses

The first research question concerned the island (in)sensitivity under clausal ellipsis. I therefore first discuss the ratings of marked remnants. The present results show a clear degradation of marked *wh*-remnants. This difference is only compatible with the view that *wh*-remnants move under ellipsis, thereby crossing an island boundary, as in (33).

- (33) * [Wie reich-en]_i ~~wird~~ ~~das~~ ~~neue~~ ~~Gesetz~~ ~~t_i~~ ~~Bürgern~~ ~~zugutekommen~~?
 how rich-ACC will the new law citizens benefit
 ‘How rich (of citizens will the new law benefit)?’

The unacceptability of (33) indicates that no ellipsis-driven repair mechanism is available to circumvent the grammatical violation in the ellipsis site. This suggests that the island violation is still present, even if left unpronounced.

In contrast, non-*wh* remnants were relatively acceptable across conditions, suggesting island insensitivity. This addresses the second research question: the present results indicate that elements, which A'-move if ellipsis had not applied, also A'-move under ellipsis, as they show island sensitivity. In contrast, non-*wh* elements do not A'-move in non-elliptical constructions. By staying in-situ, they avoid an island violation, as shown in (34).

- (34) Ja, ~~das~~ ~~neue~~ ~~Gesetz~~ ~~wird~~ ~~vor~~ ~~allem~~ ~~[sehr~~ ~~reich-en]~~ ~~Bürgern~~ ~~zugutekommen~~.
 yes the new law will above all very rich-ACC citizens benefit
 ‘Yes, (the new law will mostly benefit) very rich (citizens).’

The acceptability of configurations such as (34) indicates that ellipsis can occur around a remnant and that remnants do not necessarily have to move under ellipsis. That is, no ellipsis-driven movement needs to be stipulated.

The third research question focused on the possibility of anisomorphic interpretation. I therefore now turn to unmarked remnants. Crucially, unmarked *wh*-remnants were significantly more acceptable than their marked counterparts. One could attribute the acceptability of unmarked remnants to ellipsis repair. However, if ellipsis could repair island violations, this repair-mechanism should also be present for marked *wh*-remnants. The unacceptability of marked *wh*-remnants therefore provides strong support that ellipsis does not repair island violations. Thus, the acceptability of unmarked *wh*-remnants seems to stem from the presence of an island-evading ellipsis site, which circumvents the island violation, as in (35).

- (35) [Wie reich]_i ~~sind~~ ~~die~~ ~~Bürger~~ ~~t_i~~?
 how rich.NOM are the citizens
 ‘How rich (are the citizens)?’

I now turn to the effect of focus. The results of Experiment 2 indicate that contrastive focus generally degrades the acceptability of fragment answers. However, the significant interaction between FOCUS and CLAUSE provides crucial insights: [contrastive, non-*wh*] items received the lowest ratings across all conditions. This result is expected if one assumes that contrastive focus on the remnant forces an isomorphic ellipsis site. A contrastively-focused remnant favors an isomorphic interpretation, which requires accusative marking, as in (36).

- (36) Nein, ~~das~~ ~~neue~~ ~~Gesetz~~ ~~soll~~ ~~vor~~ ~~allem~~ ~~[arm*(-en)]~~ ~~Bürgern~~ ~~zugutekommen~~.
 no the new law will above all poor-ACC citizens benefit
 ‘No, (the new law will benefit mostly) poor (citizens).’

Crucially, there is no suitable ellipsis site available for a contrastive-focused remnant without accusative marking: the unmarked remnant is only compatible with a copular clausal ellipsis site, which is not available under contrastive focus, as in (37). If no isomorphic ellipsis site is available, the fragment answer is deemed less acceptable. The following sections evaluate the competing approaches to clausal ellipsis in the literature in light of the present findings.

- (37) #Nein, ~~die Bürger~~ ~~sind~~ [arm(*-en)].
 no the citizens are poor
 ‘No, (the citizens are) poor.’

4.2 Challenges for repair-based approaches

As outlined in Section 2.3, approaches that adopt the notion of ellipsis repair predict that island-violating and non-island configurations should receive comparable acceptability ratings under clausal ellipsis. The present results are inconsistent with this prediction.

Experiment 1 revealed a clear degradation of marked wh-remnants relative to their unmarked counterparts. Marked wh-remnants necessarily involve movement across an island boundary, as shown in (33). If ellipsis repaired island violations, these marked remnants are predicted to be fully acceptable. Instead, they received significantly lower ratings.

Crucially, the acceptability of unmarked wh-remnants cannot be attributed to island repair. If ellipsis were capable of repairing island violations, the same repair mechanism should also be available for marked wh-remnants. The fact that only the marked remnants are degraded indicates that the island violation remains present despite the ellipsis operation. The observed contrast therefore provides strong evidence that ellipsis does not repair syntactic island violations.

Further evidence comes from Experiment 2. If island repair were available, contrastive focus should not reintroduce island sensitivity. However, the results show that contrastive focus can force an isomorphic interpretation of the ellipsis site, thereby making island effects visible again. This finding is difficult to reconcile with repair-based accounts, which predict that deletion should uniformly eliminate the effects of illicit movement.

4.3 Challenges for movement-based approaches

The M&D approach assumes that all remnants escape ellipsis through movement to the left periphery before deletion applies. Under this view, ellipsis licensing uniformly relies on A'-movement, regardless of remnant type.

The present findings are problematic for this prediction. The theoretically important result is not simply that wh-remnants exhibit island sensitivity. Rather, the critical finding is the interaction between remnant type and island sensitivity. In Experiment 1, wh-remnants showed clear island sensitivity, whereas Experiment 2 revealed no comparable effect for non-wh remnants.

This asymmetry follows naturally if movement under ellipsis reflects independently attested movement properties in the non-elliptical syntax. Wh-elements undergo A'-movement in non-elliptical clauses and therefore continue to do so under ellipsis, giving rise to island sensitivity. Non-wh remnants, in contrast, do not undergo A'-movement in their non-elliptical counterparts and consequently remain in situ under ellipsis, avoiding island violations altogether.

The M&D approach cannot straightforwardly account for this contrast. If all remnants moved to escape ellipsis, island sensitivity should not depend on remnant type. Both wh- and non-wh remnants would be expected to show comparable degradation whenever movement crosses an island boundary. The absence of a difference in acceptability between marked and unmarked non-wh remnants therefore constitutes a direct challenge to movement-based approaches as the M&D approach.

Importantly, this argument does not depend on whether movement-based approaches assume isomorphic or anisomorphic ellipsis sites. Even if anisomorphic sources are permitted, the crucial prediction remains that remnants are licensed through movement. The observed diver-

gence between wh- and non-wh remnants demonstrates that movement under ellipsis cannot be a uniform requirement for ellipsis licensing. The results therefore suggest that movement under ellipsis is not triggered by ellipsis itself. Instead, movement appears to occur only when independently required by the syntax of the remnant.

4.4 Evidence supporting an in-situ, island-evading approach

The overall pattern of results is most consistent with an in-situ approach to clausal ellipsis. Under this view, remnants move only if they would independently move in the corresponding non-elliptical structure. Remnants may therefore remain in-situ, allowing ellipsis to target only the unfocused material. Furthermore, the ellipsis site need not always be isomorphic to its antecedent. When available, an anisomorphic source can be used to evade an otherwise unavoidable island violation.

The results of Experiment 1 support both components of this analysis. Marked wh-remnants exhibit island sensitivity because they involve A'-movement across the LBC. In contrast, unmarked wh-remnants are substantially more acceptable because they can be derived from an anisomorphic copular clause, as illustrated in (35). The availability of this alternative source allows the derivation to avoid the island violation altogether.

Experiment 2 further refines this picture by demonstrating that anisomorphic sources are not freely available. The crucial finding is the interaction between *CLAUSE* and *FOCUS*. Contrastive focus forces an isomorphic interpretation of the ellipsis site, in line with proposals by Barros (2013) and Griffiths (2019). Consequently, a copular clausal ellipsis site is unavailable in the [anisomorphic, contrastive] condition. Since the unmarked adjective is only compatible with a copular structure, no suitable underlying clause remains available, resulting in a decrease in acceptability.

This interaction provides evidence that anisomorphic ellipsis sites are not generally available. Instead, the findings suggest that copular clauses function as a last-resort strategy, aligning with van Craenenbroeck (2010) and Barros (2014). Anisomorphic interpretations become available to evade an island but is restricted under contrastive focus. Therefore, isomorphic interpretations seem to be the default in clausal ellipsis.

Taken together, the findings support five core claims of the in-situ approach. First, remnants move under ellipsis only if they independently undergo A'-movement in non-elliptical syntax. Second, ellipsis can apply around an in-situ remnant. Third, island violations are not repaired by ellipsis. Fourth, anisomorphic ellipsis sites are available when independent conditions permit them. Fifth, contrastive focus forces isomorphism, thereby blocking anisomorphic sources and revealing otherwise avoidable island effects.

These conclusions show that ellipsis licensing is shaped by the interaction of syntax, focus, and discourse conditions rather than by uniform movement or repair mechanisms. They also help explain the divergence from Yoshida et al. (2019), whose findings have been argued to reflect both methodological limitations and the use of weaker island configurations. By contrast, the present experiments employed the LBC and revealed clear island effects whenever movement across that island was independently required.

5. Conclusions

The experimental results strongly support an in-situ approach to clausal ellipsis. The findings demonstrate that only elements that would undergo A'-movement in non-elliptical contexts do so under ellipsis, revealing island sensitivity for wh-remnants but not for non-wh remnants. Moreover, contrastive focus on the remnant enforces an isomorphic ellipsis site, explaining why island violations persist when no anisomorphic, island-evading structure is available. These results are difficult to reconcile with classical repair-based and move-and-delete approaches and are most naturally accounted for under an in-situ analysis of clausal ellipsis. Instead, the data suggest that

ellipsis licensing is shaped by discourse factors – particularly focus and contrast – rather than by uniform syntactic movement operations. Crosslinguistic variation can thus be attributed to independent syntactic and focus-related constraints rather than to weaknesses of the in-situ model. More broadly, the results emphasize that ellipsis resolution is sensitive to discourse structure, not syntactic parallelism alone.

Abbreviations

ACC	accusative
AIC	Akaike information criterion
β	beta coefficient
CLMM	cumulative link mixed model
DAT	dative
DP	determiner phrase
F	focus
LBC	left branch condition
LBE	left branch extraction
M&D	move-and-delete
NOM	nominative
NP	noun phrase
RC	relative clause
SE	standard error
t	trace
TP	tense phrase

Data accessibility statement

The datasets, stimuli, and R-code scripts for the experiments, are publicly available via the Open Science Framework. The materials can be accessed at <https://osf.io/es76h/overview>.

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Competing interests

The author has no competing interests to declare.

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